

A commutativity spot-check with detection probability $3/8$

Counterexample to Conjecture 00000005449

ziangni-sys 4 October 2026

Claim and sampling law

Both source versions assert that a single check has detection power at least $1/2$ on a noncommutative witness. Consider the standard commutativity spot-check: draw X, Y independently and uniformly from a finite group G , and detect noncommutativity if $XY \neq YX$. Thus every ordered pair has probability $1/|G|^2$. We show that this ordinary experiment has detection probability $3/8 < 1/2$ on a genuine noncommutative group.

The actual group

Let G be the symmetry group of the square, of order eight, often denoted D_4 or D_8 depending on the indexing convention. Write its elements as

$$\{r_i : i \in \mathbb{Z}/4\mathbb{Z}\} \cup \{sr_i : i \in \mathbb{Z}/4\mathbb{Z}\}.$$

With subscripts computed modulo four, multiplication is

$$r_i r_j = r_{i+j}, \quad r_i sr_j = sr_{j-i}, \quad sr_i r_j = sr_{i+j}, \quad sr_i sr_j = r_{j-i}.$$

In particular $r_1 sr_0 = sr_3$ whereas $sr_0 r_1 = sr_1$, so the group is noncommutative.

Complete pair count

For each x , count its commuting partners using the displayed multiplication laws:

Elements x	Number	Partners per element	Ordered pairs
r_0, r_2	2	8	16
r_1, r_3	2	4	8
sr_0, sr_1, sr_2, sr_3	4	4	16

All rotations commute with rotations. A rotation r_i commutes with a reflection precisely when $2i = 0$ modulo four. Two reflections sr_i, sr_j commute precisely when $2(j - i) = 0$, so each reflection has two reflection partners and two rotation partners. The table therefore covers all elements and gives 40 commuting ordered pairs out of $8^2 = 64$. The actual detection event has $64 - 40 = 24$ pairs, so

$$\mathbb{P}(XY \neq YX) = \frac{24}{64} = \frac{3}{8} < \frac{1}{2}.$$

The possible equality $X = Y$ is included, as required for independent uniform sampling; it is not removed by conditioning. This refutes the single-check lower bound. The other clauses about independent repetitions and automorphism corrections are not needed for this disproof, and unspecified alternative algorithms are not claimed to share this sampling law.

Formal verification

Lean uses the actual Mathlib group `DihedralGroup 4`, with its group structure and eight-element finite enumeration. It proves the noncommuting witness and counts the full finite filters defined by actual equality and inequality of the two products. Each draw has rational mass $1/8$, and the pair mass is defined as the product of the draw masses. Lean verifies nonnegative pair masses and normalization, and defines detection probability by summing these weights over the actual noncommuting-pair event. The final theorem includes noncommutativity, a valid sampling distribution, the exact probability, and failure of the proposed lower bound. Finite checks use ordinary kernel-checked `decide`; no multiplication table or count is assumed and no native decision shortcut or external numerical calculation is used.